

Smart Shopper

AWS Cloud Deployment Options Report

Minimum cost vs best quality — benchmark comparison for the Smart Shopper multi-agent architecture (Kafka, NER, Playwright scrapers, Redis, MongoDB).

Project: Smart Shopper PFA | **Region recommendation:** eu-west-3 (Paris) or eu-south-2 (Spain) | **Date:** June 2026

Executive Recommendation

Best balance (recommended): Option B — ECS on EC2 (Graviton) + self-hosted Kafka + ElastiCache Valkey + MongoDB Atlas M10

Aspect	Detail
Estimated cost	~\$180–280/month (production-like beta)
Reliability	Managed cache + DB, separate compute, auto-restart
Fit	Docker images ready; no Kubernetes required yet
Avoid for now	EKS \$73/month + MSK \$450+/month until traffic justifies them

PFA / demo (lowest cost): Option A — single EC2 + docker-compose.full.yml (~\$90–150/month with Spot/Reserved Instance).

Production at scale: Option C — EKS + MSK (~\$550–900/month) when you have traffic and a Kubernetes-capable team. Existing deploy/k8s/ manifests are ready.

Your Workload Profile

Component	Resources	Notes
NER service	3–6 GB RAM, 1–2 vCPU	Hugging Face model ~1 GB; must stay warm
Scraper ×3	1–3 GB each	Playwright-heavy; 14 parallel sites
6 other agents	~512 MB each	Light Kafka consumers
Kafka	12+ topics	Most expensive managed-service mistake if mis-sized
Redis	512 MB – 1 GB	Cache + rate limits
MongoDB	512 MB – 2 GB	User memory + price watches
Minimum server	32 GB RAM, 4–8 vCPU	Or split app + data tier

Five AWS Deployment Options

Option A — Single EC2 + Docker Compose (Cheapest MVP)

Estimated cost: ~\$90–150/month

1× EC2 r7g.xlarge (4 vCPU, 32 GB) running docker-compose.full.yml. MongoDB Atlas M0 (free) optional.

Item	Monthly
EC2 r7g.xlarge (eu-west-3)	~\$184
50 GB gp3 EBS	~\$5
Data transfer	~\$5–15
MongoDB Atlas M0	\$0
Total (on-demand)	~\$194
With Spot / 1-yr RI	~\$90–120

Pros: Pros: Cheapest, identical to local dev, fast PFA demo setup.

Cons: Cons: No HA, single point of failure, manual backups.

Scores: Cost 10/10 | Reliability 3/10 | Scalability 2/10 | PFA ready 10/10

Option B — ECS on EC2 + Managed Data (RECOMMENDED)

Estimated cost: ~\$180–280/month

2× EC2 r7g.large ECS workers. Self-hosted Kafka on t4g.medium OR MSK kafka.t3.small. ElastiCache Valkey cache.t4g.small. MongoDB Atlas M10. ECR for images.

Item	Monthly
2× r7g.large ECS workers	~\$184
1× t4g.medium Kafka (self-hosted)	~\$30
ElastiCache cache.t4g.small (Valkey)	~\$24
MongoDB Atlas M10	~\$57
ECR + CloudWatch + secrets	~\$10–20
Total	~\$305
With Spot + 1-yr RI	~\$180–220

Pros: Pros: Best cost/quality ratio; managed Redis + Mongo; Dockerfiles work as-is.

Cons: Cons: You manage Kafka (or ~\$100 for minimal MSK).

Scores: Cost 9/10 | Reliability 7/10 | Scalability 7/10 | Production 8/10

Option C — EKS + MSK + Managed Services (Production)

Estimated cost: ~\$550–900/month

EKS cluster using existing deploy/k8s/ manifests. MSK kafka.t3.small × 3. ElastiCache + Atlas M10. HPA for scrapers (hpa.yaml).

Item	Monthly
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EKS control plane	\$73
2× m7g.xlarge nodes	~\$278
MSK kafka.t3.small × 3	~\$100
ElastiCache cache.t4g.small	~\$24
MongoDB Atlas M10	~\$57
NAT Gateway + ALB + logs	~\$52–85
Total	~\$584–627
Under HPA peak load	~\$800–1,200

Pros: Pros: Matches K8s manifests, HPA, multi-AZ capable, production-grade.

Cons: Cons: \$73/month before compute; needs Kubernetes expertise (3–6 months learning).

Scores: Cost 4/10 | Reliability 10/10 | Scalability 10/10 | K8s fit 10/10

Option D — ECS Fargate (Simplest Ops)

Estimated cost: ~\$350–500/month

All 13 containers on Fargate (serverless). MSK or self-hosted Kafka. ElastiCache + Atlas M10.

Item	Monthly
NER (2 vCPU, 4 GB)	~\$60
Scraper ×3 (1 vCPU, 2 GB)	~\$90
7 other agents	~\$70
Kafka + Redis + Mongo	+\$130–280
Total	~\$350–500

Pros: Pros: No EC2 patching; auto-scaling per service.

Cons: Cons: 40–60% more expensive than ECS on EC2 for 24/7 workloads.

Scores: Cost 6/10 | Reliability 8/10 | Ops simplicity 9/10

Option E — MSK Serverless (AVOID for MVP)

Estimated cost: ~\$730–860+/month

MSK Serverless has ~\$0.75/hour cluster fee = ~\$558/month minimum even with no traffic.

Item	Monthly
MSK Serverless (cluster only)	~\$558
Partitions + data	+\$20–100
EC2 for agents	+\$150–200
Total	~\$730–860+

Pros: Do NOT use for PFA/MVP. Use self-hosted Kafka or MSK Provisioned kafka.t3.small instead.

Cons: No HA benefit over cheaper alternatives at this price point.

Scores: Cost 2/10 | Worst choice for low-traffic MVP

Benchmark Matrix (1 = poor, 10 = excellent)

Criteria	A: EC2	B: ECS	C: EKS	D: Fargate	E: MSK SL
Lowest cost	10	9	4	6	2
Reliability / HA	3	7	10	8	8
Scalability	2	7	10	8	9
Ops simplicity	8	7	3	9	6
Fits Docker setup	10	9	8	9	7
Fits K8s YAML	2	4	10	4	6
Playwright scraping	7	8	9	7	7
NER isolation	6	8	9	8	7
PFA / demo ready	10	9	5	8	3
Production ready	3	8	10	8	7
Monthly cost (\$)	90–194	180–305	584–900	350–500	730–860
Weighted score	6.2	8.1	7.4	7.6	4.8

Managed Services — What to Pick

Service	Cheapest good choice	Avoid
Kafka	Self-hosted t4g.medium (~\$30) or MSK kafka.t3.small (~\$33 dev)	MSK Serverless (~\$558/mo base)
Redis	ElastiCache Valkey cache.t4g.small (~\$24, 20% cheaper)	Serverless for steady cache
MongoDB	Atlas M0 (demo) → M10 (~\$57 prod)	DocumentDB (different API)
LLM	Groq API (pay per token)	SageMaker / GPU EC2 (\$500+/mo)
Secrets	AWS Secrets Manager + SSM	Plain text .env on EC2
Images	ECR	Docker Hub rate limits
Monitoring	CloudWatch + /metrics endpoint	Paid Grafana Cloud at start

3-Phase Rollout Plan

Phase	When	Architecture	Est. cost/mo
1 – PFA demo	< 100 users	1× EC2 + docker-compose.full.yml + Atlas M0	\$90–150
2 – Beta	First real users	ECS + EC2 Spot + Valkey + Atlas M10 + self-hosted Kafka	\$180–280
3 – Production	1K+ users, need HPA	EKS + MSK + Valkey + Atlas M10 (deploy/k8s/)	\$550–900

What NOT to Do

Mistake	Why it hurts
MSK Serverless for MVP	~\$558/month minimum before any messages
EKS on day 1 without K8s experience	+\$73/mo + high ops time; delays PFA

Fargate for all 13 services 24/7	40–60% more than ECS on EC2
GPU EC2 for LLM	Groq API is cheaper (LLM_PROVIDER=groq)
Playwright on all 14 sites at max concurrency	RAM spikes; keep SCRAPE_PLAYWRIGHT_PROVIDERS=avito
us-east-1 for Moroccan users	Higher latency; use eu-west-3 (Paris)

External Costs (Not AWS)

Item	Cost
Groq LLM API	~\$0–20/mo at MVP volume
Telegram Bot	Free
Hugging Face NER model	Free (cached in NER Docker image)
Route 53 domain	~\$0.50/mo if needed

Final Answer

Goal	Best option
Absolute minimum cost	Option A — EC2 + Docker Compose (~\$90–150/mo)
Best cost + quality (recommended)	Option B — ECS on EC2 + Valkey + Atlas M10 (~\$180–280/mo)
Best quality at scale	Option C — EKS + MSK (~\$550–900/mo)

For Smart Shopper as a PFA project moving toward real users, Option B is the sweet spot: professional enough for demos and early production, 3–4× cheaper than full EKS+MSK, and migration to existing Kubernetes manifests (deploy/k8s/) requires no application code changes.

Generated for Smart Shopper — ENIAD IA PFA Project. Pricing estimates based on AWS public rates (eu-west-3, June 2026). Verify with AWS Pricing Calculator before deployment.